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Residential mobility in Lausanne: predicting departures and finding trajectory types

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Abstract

Residential mobility shapes how a city plans housing and services, yet the administrative registers that record it are built for management, not analysis. This project turns eleven years of monthly journals from the Lausanne population register (863,577 consolidated events) and a 2014 snapshot into 493,912 residential episodes for 310,746 persons, with a measured reconstruction uncertainty of 1.0% of transitions. Two questions are addressed. A supervised model predicts eventual departure from entry-time attributes alone, reaching an AUC of 0.782; trajectory features are deliberately excluded, and a contaminated model (AUC 0.947) demonstrates why: in a censored window, the unfolding trajectory encodes the outcome itself. An unsupervised analysis yields five trajectory types and six entry profiles whose departure rates, never used in the clustering, span 80% to 31% and 66% to 34%. Crossing the profiles with geography explains the city's most anomalous district figure. Departure is thus structurally underdetermined at entry: a meaningful part is inscribed in the starting conditions, but not the majority.

This report uses confidential register data; results are published at aggregate level only.

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1 Introduction

People move a lot. They arrive in a city, change flats, sometimes leave for another town, and sometimes come back years later. For a city like Lausanne, knowing how residents move matters for many practical reasons: planning housing, schools, and public services all depend on understanding who stays, who leaves, and when [1].

This project studies residential mobility using the population register of Lausanne. The register records every change that happens to a resident: arrivals, departures, moves inside the city, births, and deaths. My goal is to turn this administrative record into something useful for analysis, and then to answer two questions.

Residential mobility is best understood as a life-course process, in which moves follow and trigger family and work transitions [1]. Most quantitative evidence, in Switzerland as elsewhere, comes from surveys or from linked administrative data at the national level; Lacroix, Gagnon and Wanner [2], for instance, use Swiss administrative records to relate family changes to residential moves. This project works one level below: the full event grain of a single municipal register, which trades national coverage for a complete, dated record of every administrative change affecting every resident of one city.

The first question is a prediction task: can we tell, from what is known about a person at the moment they settle in Lausanne, whether that person will eventually leave the city? This framing is deliberate. A resident's unfolding history cannot be used for this prediction without cheating, because the history encodes the outcome itself; I demonstrate this point explicitly, and it shapes the whole supervised analysis. The second question is a grouping task: are there typical patterns in the way people move through the city over time, and can we describe them? The first question is a supervised learning problem (we know the answer for past residents and try to learn from it). The second is an unsupervised one (we look for structure without knowing the groups in advance).

The main difficulty of the project is not the modelling itself. It is the data. The register was not built for research. It is a log of administrative actions, written in the order the clerks entered them, not in the order events happened in real life. Before any analysis is possible, the data have to be cleaned, understood, and rebuilt into proper resident histories. A large part of this report describes that work, because it is where most of the effort went and where the most important decisions were made.

The report is organised as follows. Section 2 describes the data, how resident histories were rebuilt from it, and what this reconstruction is worth. Section 3 presents the exploratory analysis. Section 4 covers the methods. Section 5 presents the results, and Section 6 discusses their significance and limits. Section 7 concludes and sketches possible extensions. The full pipeline is available as eight documented notebooks in the project repository.

2 Data

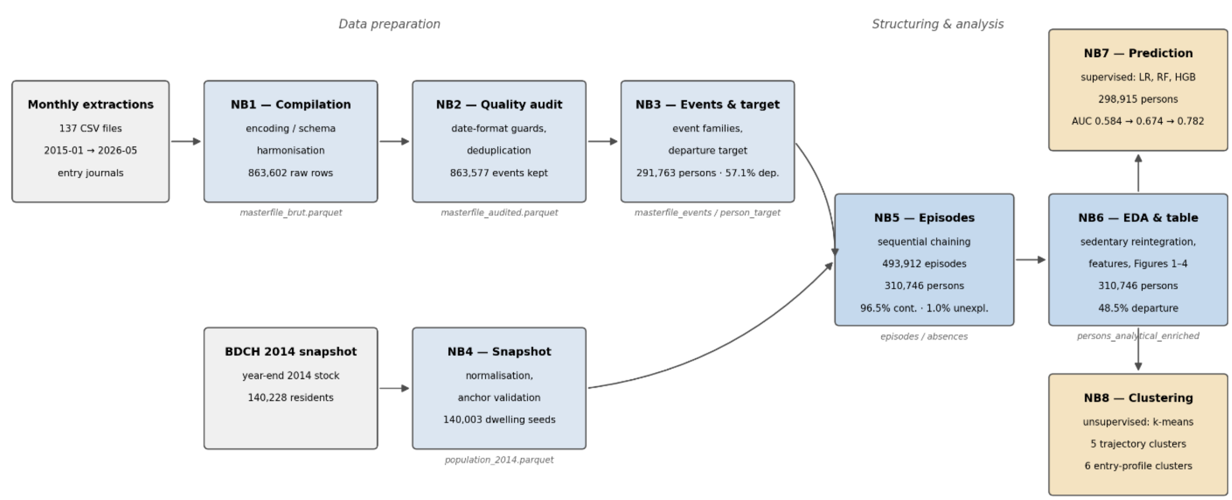
2.1 Sources

All analyses rest on the population register of the City of Lausanne (BDCH, Base de données du contrôle des habitants). I use two extractions from this register. The first is a set of 137 monthly files covering January 2015 to May 2026. Each file is a journal of register entries: one row per mutation recorded during that month, such as an arrival, a departure, a move within the city, a birth, a death, or an administrative correction. A recorded entry carries two dates, the date on which the clerk entered it and the date on which the event took effect. The two can differ by months or even years, because the register is corrected retroactively. The second extraction is a snapshot of the resident population at the end of 2014, with 140,228 individuals. It anchors the analysis: it tells me who was already living in Lausanne when the observation window opens.

A data dictionnary of the BDCH source files is available in Annex A.

A third file maps each building identifier (EGID, from the federal register of buildings and dwellings, RegBL [4]) to one of the 18 statistical districts of Lausanne. Individuals are identified throughout by a pseudonymised key derived from the register identifier; no name, address string, or birth date leaves the processing environment. The data are confidential and were processed entirely within the statistical office. **Figure 1** summarises the processing pipeline; the remainder of this section walks through it.

Figure 1 : Data pipeline



2.2 From entry journals to a consolidated event table

Concatenating the monthly files, after harmonising encodings and column schemas, yields a master table of 863,602 rows and 70 columns (notebook 1). This table is not yet a table of events. The journals record administrative entries, and the same real-world move can appear several times: as an original entry, as a cancellation (the register's Suppression codes), and again as a corrected re-entry. Notebook 2 therefore consolidates the journals in two passes. First, each cancellation is matched to the prior entry it cancels, using a preference order of matching rules: a crossed-address lock (the cancellation carries the cancelled state in its mutation columns and the restored state in its normal columns), then a shared effective date, then the most recent unmatched candidate as a fallback. Second, re-entered events are deduplicated with a last-write-wins rule keyed on the person, the mutation code, the effective date, and the destination building; the key deliberately excludes the dwelling and household identifiers, because those are precisely what re-entries correct. Nothing is deleted at any point: superseded rows are flagged and counted, so every consolidation decision remains auditable, in line with register-statistics practice [3]. Exact duplicate rows are rare (25 rows). The consolidated table contains 863,577 events for 291,763 distinct persons.

2.3 Data quality

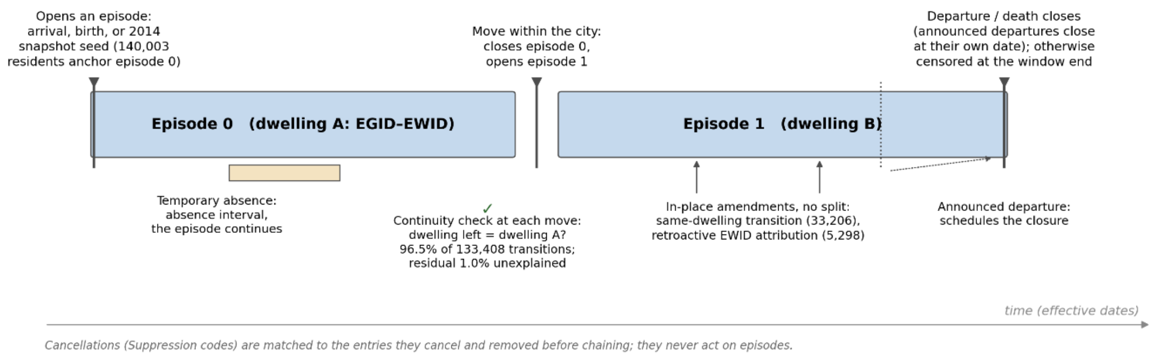
Three properties of the register affect the analysis directly and shaped the pipeline.The first is the reliability of dates. Every event carries the date on which it took effect, but this date is typed by a clerk and is sometimes wrong, and often corrected after the fact: the register contains retroactive corrections whose effective date precedes events that were already recorded, and a small number of implausible values, such as entry dates lying decades in the future. Erroneous dates matter more here than they would in a cross-sectional analysis, because the episode reconstruction of Section 2.4 orders each person's events by effective date: a single wrong date can reorder a trajectory and break the chain. The pipeline therefore treats dates defensively. Implausible values are counted and set aside; retroactive corrections that would make an episode start before the previous one ends are clipped and flagged rather than silently accepted (the counts are reported in notebook 5 §4(a)); and the residual effect of undetected date errors is one of the components of the 1.0% of unexplained transition ruptures reported in Section 2.4, which I treat as the pipeline's uncertainty on trajectory reconstruction.

The second is the completeness of the dwelling identifier. The building identifier (EGID) is reliably filled, but the dwelling within the building (EWID) is frequently missing or set to zero, especially at arrival, and is often attributed retroactively once the register catches up. The chaining rules of Section 2.4 are designed around this: a retroactive attribution of the dwelling identifier amends the open episode instead of splitting it (5,298 cases), and a renumbering of dwellings within the same building is recognised as a documented rupture cause (299 cases) rather than mistaken for a move. Dwelling attributes such as rooms and surface share the same incompleteness; their treatment by imputation with missingness flags is described in Section 2.5.

The third is geography. Districts are attached by mapping the building identifier to the 18 statistical districts of the city. The mapping table itself is imperfect: it contains 34 duplicated building entries, resolved by keeping the first occurrence. On the final corpus every remaining building matched a district, but 65 persons could not be assigned to any district and are excluded from district-level figures, and two technical codes are treated separately throughout: code 90 (zones foraines, dominated by collective households) and code 99 (unknown address).

2.4 Reconstructing residential episodes

Figure 2 : Register mutation types and their effect on residential episodes



The analytical unit of this project is the residential episode: a continuous stay of one person in one dwelling, identified by the building and dwelling pair (EGID-EWID). Episodes are not observed directly; notebook 5 reconstructs them by chaining the consolidated events of each person in effective-date order.

The chain starts from the 2014 snapshot for the individuals present then: 140,003 of the 140,228 snapshot residents open a first episode in their recorded dwelling (the remainder are secondary residences, administratively wiped files, or non-dwelling situations, and are excluded). Events then open, close, or amend episodes according to their role. Two amendment rules matter for data quality. A transition that lands in the dwelling the person already occupies is not a move: it encodes a household recomposition or a correction, and it amends the open episode instead of splitting it (33,206 cases). Similarly, when the register retroactively refines a building-level location into a full dwelling identifier, the open episode is amended in place (5,298 cases). Announced departures carry their own effective date and close the episode at that date; deaths and temporary absences are handled by dedicated rules.

The result is 493,912 episodes for 310,746 persons, plus 4,496 documented absence intervals. The reconstruction is validated in three ways. First, episodes of one person may not overlap and may not have negative durations; violations produced by retroactive corrections are clipped and counted rather than hidden. Second, at each move the dwelling the person leaves should match the dwelling of the previous episode. This holds for 96.5% of the 133,408 observed transitions. The 4,723 exceptions are classified by known cause (divergence between the snapshot dwelling and the first observed move, cancelled events sitting between two episodes, register renumbering of dwellings within a building, episodes that themselves started blind), which leaves 1,308 unexplained ruptures, or 1.0% of transitions. I treat this figure as the pipeline's uncertainty on trajectory reconstruction. Third, the ending of each person's last episode is cross-tabulated against the person-level departure status established independently in notebook 3; the two views agree for 96% of individuals, and the disagreements concentrate where they should, on non-confirmed departures.

2.5 Analytical table and modelling scope

Notebook 6 assembles the person-level analytical table. Its population is the 310,746 individuals with at least one reconstructed episode. This figure reconciles exactly with the event table: 291,763 persons generate events, of whom 20,870 have no reconstructible stay and are excluded, while 39,853 snapshot residents never generate a single chainable event and are reintegrated as certain non-departures (291,763 - 20,870 + 39,853 = 310,746). The 20,870 exclusions are almost entirely by construction: 13,690 individuals whose whole history is in secondary residence, 4,732 whose history is dominated by cancellations, 1,407 administratively wiped files, and 85 single events that cannot form an interval, leaving a residual of 956 individuals (0.3% of the population) excluded without a fully documented cause.

Table 1 : Reconciliation of the analysis perimeters, from the consolidated event table to the supervised modelling population.

Population	n
Persons in the consolidated event table (notebook 3)	291,763
Excluded: no reconstructible stay	− 20,870
entirely in secondary residence	13,690
history dominated by cancellations	4,732
administratively wiped files	1,407
single event, no interval possible	85
residual, cause not fully documented	956
Reintegrated: sedentary snapshot residents (no chainable event)	+ 39,853
Analytical population (notebook 6)	310,746

Excluded: deaths	− 11,651
Excluded: announced departures not yet effective	− 180
Supervised modelling population (notebook 7)	298,915

The table carries the departure target (48.5% of persons left Lausanne during the window; 51.1% are right-censored), trajectory descriptors (number of episodes, durations, absences), and entry-time attributes: entry year and entry type, age at entry, sex, nationality group, permit group, provenance group, household type and size, dwelling rooms and surface, and the district of the first dwelling. Missing numeric attributes are imputed with medians computed on the training split only, and each imputed variable carries a missingness flag. **Table 2** describes the composition of the analytical population and the raw departure rate of each group, which anticipates several results of Sections 3 and 5.

Table 2 : Composition of the analytical population (310,746 persons) and raw departure rate per group.

Category	n	Share (%)	Departure rate (%)
Sex			
Male	156,372	50.3	50.3
Female	154,364	49.7	46.7
Indeterminate	10	<0.1	–
Nationality group			
Swiss	127,325	41.0	37.5
EU / EFTA	118,662	38.2	58.0
Third country	64,759	20.8	52.9
Permit group			
Swiss (no permit)	127,325	41.0	37.5
P	91,105	29.3	65.0
C (settlement)	42,730	13.8	37.0
B (residence)	39,132	12.6	57.2
Asylum (F, N)	6,837	2.2	46.0
L (short duration)	2,718	0.9	73.4
Other	899	0.3	58.7
Household type			
Private	284,497	91.6	46.4
Collective	26,051	8.4	71.5
Administrative	185	<0.1	45.4
Out of municipality	13	<0.1	–
Entry type			
Arrival (2015 or later)	136,639	44.0	58.6
2014 snapshot stock	140,000	45.1	39.8

Category	n	Share (%)	Departure rate (%)
Birth in Lausanne	18,420	5.9	33.5
Stay observed mid-course (reprise)	15,684	5.0	56.4
Transition	3	<0.1	–
Age at entry			
0–17	58,770	18.9	37.7
18–29	105,733	34.0	65.4
30–44	77,507	24.9	51.0
45–64	44,742	14.4	35.0
65+	23,988	7.7	17.8

Departure rates are omitted for categories with fewer than 100 persons.

The supervised analysis of notebook 7 uses a restricted perimeter: deaths (11,651) and departures announced but not yet effective (180) are excluded, leaving 298,915 individuals with a 50.4% departure rate, split 75/25 into training and test sets with stratification on the target.

2.6 Known limitations

Four limitations should be kept in mind. First, trajectories of the 2014 stock are left-truncated: their durations are measured from the recorded arrival date, which can lie decades before the observation window, and only individuals still present in 2014 are observed, which mechanically inflates their apparent stability. Second, all durations are right-censored at 31 May 2026. Third, the 1.0% of unexplained ruptures described above propagates as an irreducible uncertainty on trajectory-based quantities. Fourth, secondary residences and administratively wiped files are outside the scope by design, so the results describe the population with a reconstructible main residence in Lausanne. In the survival analysis of Section 3, deaths are treated as censoring rather than as a competing risk, and 37 individuals with zero recorded duration are excluded; both choices are discussed there.

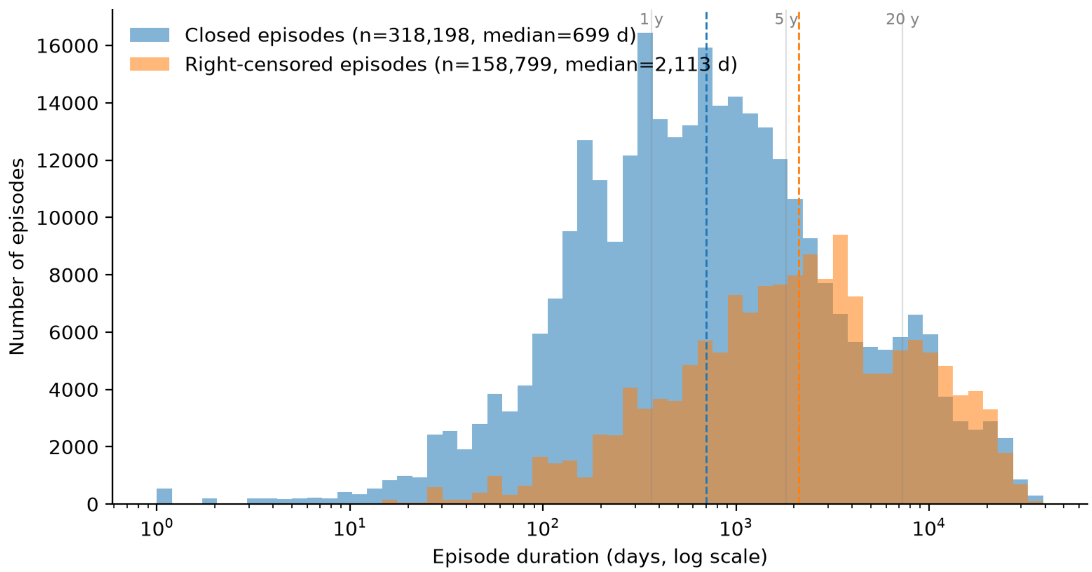
3 Exploratory data analysis

This section describes the reconstructed trajectories before any modelling. Four figures carry the analysis: the distribution of episode durations (**Figure 3**), survival curves for the time to departure (**Figure 4**), mobility within the city (**Figure 5**), and the geography of departure (**Figure 6**). The composition of the population itself was described in **Table 2**.

3.1 How long does a stay in one dwelling last?

Figure 3 shows the distribution of episode durations on a logarithmic scale, separating closed episodes from right-censored ones. The median closed episode lasts 699 days, slightly under two years. Right-censored episodes, those still open at the end of the window, have a much higher median of 2,113 days: unsurprisingly, the dwellings people have not yet left are the dwellings people stay in. The distribution is wide, spanning from a few days to several decades, which is a first hint that the population mixes very different mobility regimes. A further 16,915 episodes have a recorded duration of zero days; they are administrative chains of same-day events (for instance an arrival immediately corrected) and are excluded from the plot but kept in the episode table.

FIGURE 3 : RESIDENTIAL EPISODE DURATIONS IN LAUSANNE (2015-2026)

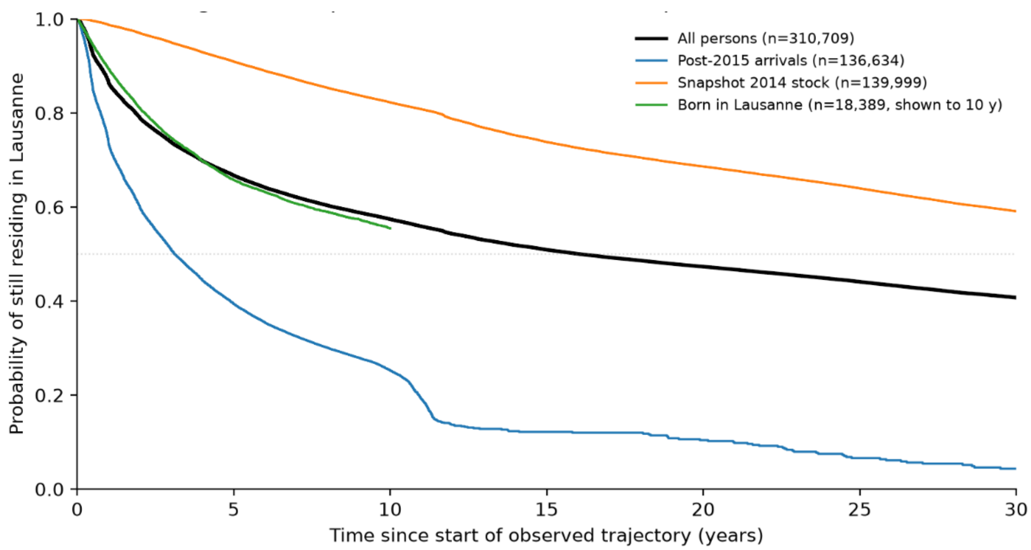


3.2 How long until people leave Lausanne?

FIGURE 4 presents Kaplan-Meier survival curves [5] for the time between the start of a trajectory and the departure from Lausanne, with deaths treated as censoring and 37 persons with zero recorded duration excluded. Over the whole population, the median survival is 16.0 years: half of the residents observed in the window are gone within sixteen years of the start of their trajectory. This overall figure, however, averages three groups that the stratified curves show to be entirely different.

Residents who arrived in 2015 or later are the only stratum measured without bias, since their trajectories are observed from the true beginning. Their median survival is 3.1 years: half of the newcomers have left the city within about three years, and the curve falls fastest in the first two. The 2014 snapshot stock shows a nominal median of 49.2 years, but this figure should not be read as a survival time: the stratum is left-truncated, since only people still present at the end of 2014 are observed, and long-settled residents are mechanically over-represented.

FIGURE 4 : KAPLAN-MEIER SURVIVAL : TIME TO DEPARTURE FROM LAUSANNE



The curve of residents born in Lausanne is displayed only to ten years, because births are observed from 2015 onward and the tail beyond that point rests on a handful of retroactively registered births. The figure therefore does double duty: it estimates survival where the data allow it, and it makes the two symmetric biases of the observation window visible, left truncation for the old stock and right-edge effects for the birth cohort.

3.3 Mobility within the city

FIGURE 5 : RESIDENTIAL EPISODES PER PERSON AND DEPARTURE RATE

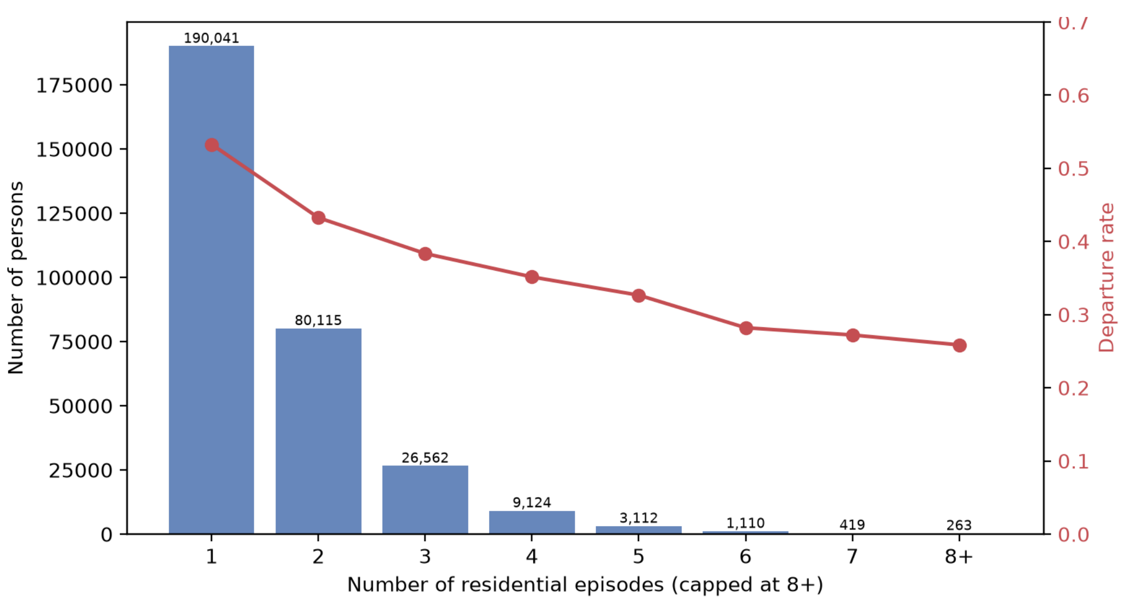
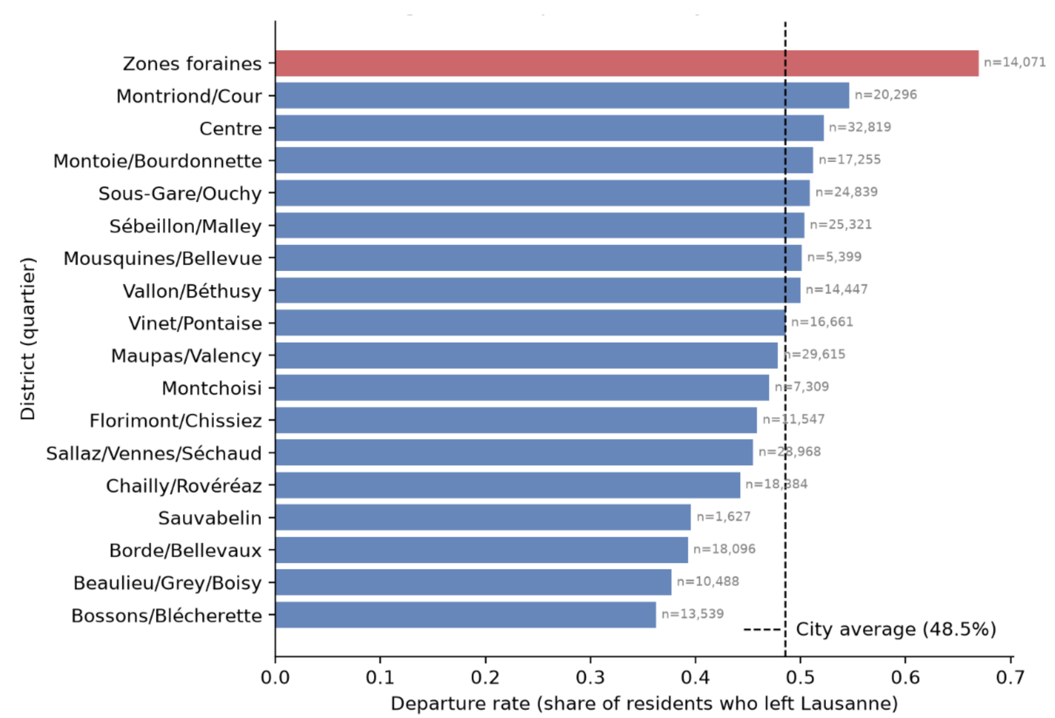


FIGURE 5 turns to moves inside Lausanne. Mobility within the city is limited: 61.2% of the analytical population occupies a single dwelling over the whole observed trajectory, and only 13.1% occupy three or more. More interesting is the red line of the figure: the departure rate falls steadily with the number of residential episodes, from 53.3% among single-episode residents to 30.8% among those with five or more. Internal mobility is thus a signal of attachment rather than of instability: people who move within Lausanne tend to stay in Lausanne. Part of this association is mechanical, since accumulating episodes requires remaining in the city long enough to move, and this exposure effect is exactly the kind of information that makes trajectory descriptors unusable for honest prediction, as Section 4 develops.

3.4 The geography of departure

FIGURE 6 maps the departure rate onto the 18 statistical districts, using the district of each person's first dwelling. The spread is wide: from about 36% in the peripheral residential districts (Bossons/Blécherette, Beaulieu/Grey/Boisy) to 67% in the Zones foraines, with the city centre and the districts around the station above the 48.5% average.

FIGURE 6 : DEPARTURE RATE BY LAUSANNE DISTRICT



Note: 65 persons with unknown address (code 99) and districts with n < 100 are excluded. 'Zones foraines' (90) is a technical district dominated by collective househc

The Zones foraines figure is less geographical than institutional: this technical district concentrates collective households, whose departure rate is 71.5% (**TABLE 2**), largely temporary residents of institutions and residences (e.g. EHL Hospitality Business School). The centre-versushills gradient in the remaining districts, in contrast, is a genuinely residential pattern, and Section 5 shows that it aligns with where different entry profiles settle.

3.5 What the exploration sets up

Three facts from this section shape the modelling choices. First, departure is concentrated early in the trajectory: whoever survives the first years of presence is likely to stay, which makes the moment of entry the natural point of prediction. Second, the population mixes regimes, from short-stay transients to residents settled for decades, which motivates the search for a typology rather than a single average description. Third, trajectory descriptors such as the number of episodes or the observed duration are saturated with exposure information: they describe the outcome as much as the person, and the supervised analysis must be designed so that they cannot leak into the features.

4 Methods

Both analyses run on the perimeters defined in Section 2.5 and were implemented in Python with scikit-learn [7] and pandas [8]; survival curves use lifelines [6]. All random operations are seeded, and the notebooks that produce every number and figure of this report are available in the project repository.

4.1 Supervised analysis: predicting departure at entry

The task is a binary classification: for each of the 298,915 persons in the modelling perimeter, predict whether the person left Lausanne during the observation window (50.4% did). The population is split once into a training set (224,186 persons) and a held-out test set (74,729), with a 75/25 stratified draw on the target and a fixed seed; the split is persisted to a file on first run and reused afterwards, so that every model in the project faces the exact same test set. Model comparison and tuning rely on five-fold cross-validation within the training set only; the test set is touched once per model, for the final figures reported in Section 5.

Features are restricted to what is known about a person at the moment their trajectory starts: entry year and entry type, age at entry, sex, nationality group, permit group, provenance group, household type and size, dwelling rooms and surface, and the district of the first dwelling. Categorical variables are one-hot encoded, which yields 54 columns. Missing numeric values are imputed with medians computed on the training set only, and each imputed variable carries a missingness indicator, so that the model can learn from the absence itself.

The restriction to entry-time features is the central methodological decision of this analysis, and I demonstrate rather than assume it. Trajectory descriptors, such as the number of episodes, the observed duration, or the number of absences, are excellent predictors for a bad reason: in a censored window, a short observed duration is often the departure itself, seen from another angle. To make the point measurable, one deliberately contaminated model is trained with these descriptors included. Its performance, reported in Section 5, is spectacular and meaningless, and it serves as an upper bound on how misleading a naive design would be. A second, subtler channel is quantified the same way: the entry year encodes exposure time (someone who arrived in 2024 has had little time to leave), so the honest model is also refitted without it, and the difference measures how much of its performance is calendar exposure rather than personal attributes.

Three model families are compared under this protocol: a logistic regression on entry type alone, as a floor; a logistic regression on the full entry-time features, which is the interpretable reference model; and two tree ensembles, a random forest and gradient-boosted trees (HistGradientBoosting), which capture interactions and non-linearities at the price of interpretability. The primary metric is the area under the ROC curve (AUC), well suited to the near-balanced target; accuracy, precision, recall and F1 are reported alongside. Coefficients of the logistic models are read on standardised features, and the error structure of the best model is examined on the test set to identify where it fails.

4.2 Unsupervised analysis: two typologies

Two complementary clusterings are built with k-means on the same 298,915 persons. The first describes trajectories: number of episodes, median episode duration, mobility intensity (episodes per observed year), age at entry, and household size. Because k-means is sensitive to outliers and scale, durations and household size enter in logarithm, mobility intensity is winsorised at its 99th percentile, and all features are standardised. The second typology deliberately ignores trajectories and describes people as they arrive, through twelve entry-time attributes: ten binary indicators (female; EU/EFTA nationality; third-country nationality; foreign provenance; recent arrival in Switzerland; temporary permit; married; family with children; single-parent household; recent birth in the household) together with age at entry and household size, the latter in logarithm. Keeping the two feature spaces disjoint from the outcome is what allows the departure rate of each cluster to serve as an external validation rather than a tautology.

The number of clusters is selected with three instruments and one priority rule. Silhouette scores [9] are computed on a random sample of 20,000 persons, since the exact computation is quadratic; the inertia elbow gives a coarse range; and the deciding criterion is stability: the partition is refitted under five different random seeds and compared to the reference with the adjusted Rand index (ARI) [10]. Every fit uses thirty random restarts (n_init = 30). This last setting matters more than it seems: with the default number of restarts, an early run produced a stability verdict that reversed after a marginal data correction, showing that the diagnostic was partly measuring initialisation luck. Stability diagnostics are estimators too [11], with their own variance, and they must be computed carefully enough to be trusted.

The retained partitions are k = 5 for trajectories and k = 6 for entry profiles; the selection evidence is reported in Section 5. Each cluster is profiled on its own features (medians on the original scale) and then characterised a posteriori on variables kept out of the clustering, departure rate first. Cluster numbering is arbitrary and permutes between runs, so clusters are named through distinctive signatures (for instance, the cluster with the highest share of a given flag) rather than through their numeric labels; the naming survived every rerun of the pipeline unchanged, which validates the mechanism.

5 Results

5.1 Predicting departure at entry

Table 3 compares the models on the held-out test set. The progression tells the story better than any single number. Knowing only how a person entered the register (arrival, birth, snapshot stock) already beats chance, with an AUC of 0.584. Adding the full set of entry-time attributes brings the interpretable logistic regression to 0.674. Letting tree ensembles exploit interactions and non-linearities adds another eight points, to 0.775 for the random forest and 0.782 for gradient boosting. Five-fold cross-validation on the training set gives the same figures to within 0.001, and the ranking is identical, so the comparison is not an artefact of one particular split.

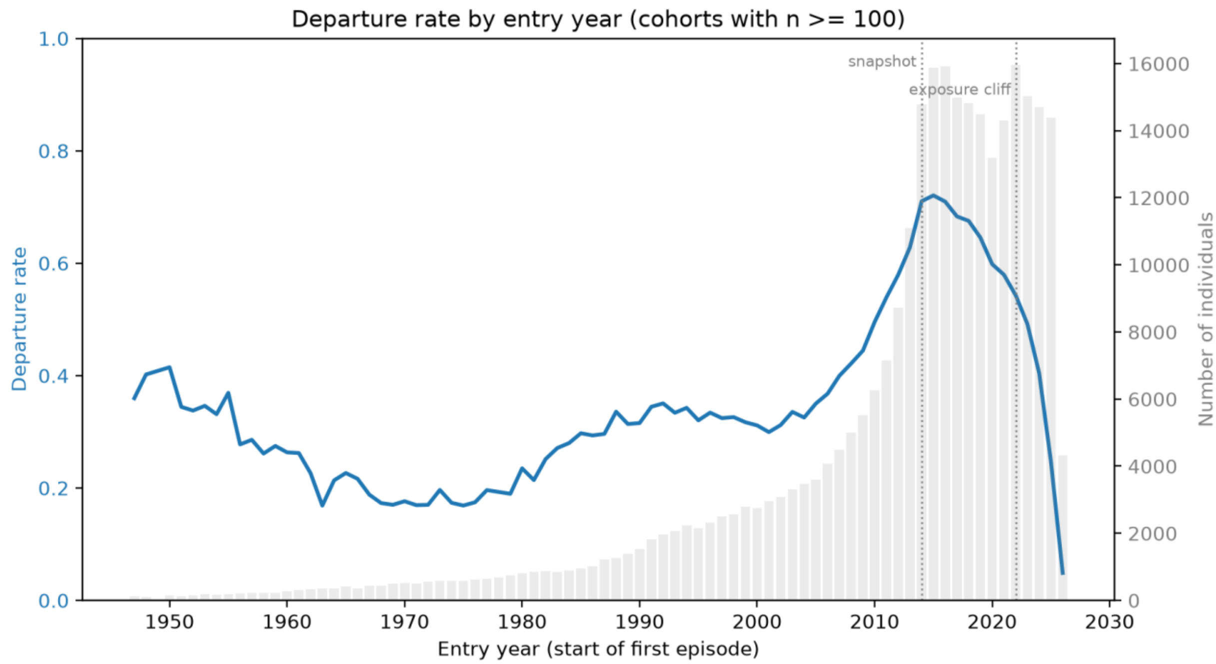
Table 3 : Departure prediction on the held-out test set (74,729 persons).

Model	Accuracy	Precision	Recall	F1	AUC
Entry type only (logistic)	0.584	0.583	0.618	0.600	0.584
Entry-time features (logistic)	0.627	0.634	0.613	0.624	0.674
Random forest	0.699	0.701	0.701	0.701	0.775
Gradient boosting (HGB)	0.705	0.703	0.718	0.711	0.782
Contaminated: trajectory features included	0.898	0.955	0.837	0.892	0.947

The last row is the deliberately contaminated model and is reported as a leakage bound, not as a result.

The last row of **Table 3** is the deliberately contaminated model of Section 4.1. With trajectory descriptors included, AUC jumps to 0.947. This number is spectacular and meaningless: in a censored window, a short observed duration largely is the departure, so the model is shown the answer. The nine-point gap between 0.782 and 0.674, in contrast, is legitimate signal, and the second bound of Section 4.1 decomposes part of it: refitted without the entry year, the gradient boosting model falls to 0.730, so about five of those points reflect calendar exposure (early entrants have had more time to leave) rather than personal attributes.

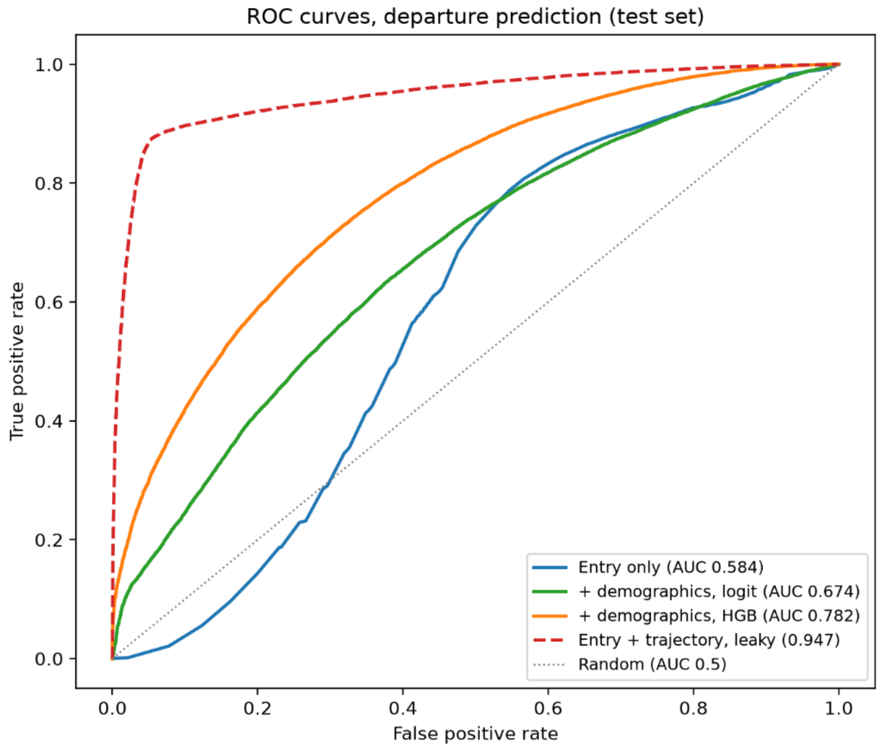
FIGURE 7 : DEPARTURE RATE BY ENTRY YEAR



What predicts departure is consistent across the logistic coefficients, the tree-based importances, and the raw rates of **TABLE 2** : temporary permits, arrival as an adult between 18 and 29, foreign provenance and collective households push towards departure; Swiss nationality, entry by birth, entry at older ages and larger dwellings push towards staying. The cohort curve of adds the time dimension: departure rates by entry year are stable around 20 to 40% for the pre-2010 stock, rise for recent cohorts, and then collapse for the most recent ones, which simply have not had time to leave yet; the two annotated breaks (the 2014 snapshot boundary and the exposure cliff) mark where interpretation must stop.

Error analysis on the test set locates the model's blind spots rather than just measuring them. Among the residents most confidently predicted to leave who nevertheless stayed, holders of temporary permits living in private households are over-represented: the model reads the permit, but integration into a private household retains people in ways entry-time data cannot see. Conversely, the surprising leavers are concentrated among older residents of the 2014 stock, whose departures (often to neighbouring municipalities or care settings) have no entry-time signature at all.

FIGURE 8 : ROC (RECEIVER OPERATING CHARACTERISTIC) CURVES AND DEPARTURE PREDICTION



5.2 A typology of trajectories

For the trajectory clustering, the silhouette marginally peaks at k = 6 (0.299 against 0.293 for k = 5), and at thirty restarts both partitions are highly reproducible (minimum ARI 0.997 for k = 5, 0.991 for k = 6). Stability therefore does not discriminate, and the decision falls back on the first criterion of Section 4.2: at k = 6 the additional cluster is an artifact group covering 0.9% of the population, residents of very large collective households, not a trajectory type. The retained partition has five clusters, summarised in **TABLE 4** with their departure rates, which were not used in the clustering.

TABLE 4 : TRAJECTORY CLUSTERS (K = 5), ORDERED BY DEPARTURE RATE

Trajectory cluster	n	Share	Median signature	Departure
Short-stay transients	37,695	12.6%	1 episode of 121 days	80.0%
Standard arrivals	94,181	31.5%	1 episode of 760 days	63.7%
Stock families / children	62,899	21.0%	entry at age 8, household of 4	42.1%
Internal movers	36,768	12.3%	3 episodes of 624 days	36.4%
Long-settled residents	67,372	22.5%	1 episode of 8,063 days	30.7%

Signatures are medians on the original scale; departure rates are external to the clustering.

The gradient is steep and coherent: four in five short-stay transients leave, against fewer than one in three long-settled residents. Between these poles, the typology separates two forms of stability, the immobile long-settled and the internal movers whose three successive dwellings confirm the attachment signal of **Figure 5**, and two forms of arrival, the standard adult arrival and the children of the 2014 stock.

5.3 A typology of entry profiles

For the entry profiles, silhouette is nearly flat between k = 5 and k = 7 (0.255, 0.250, 0.254) and all three candidates are reproducible at thirty restarts (minimum ARI 0.999, 0.952, 0.948). The retained k = 6 is the partition in which every cluster is a recognisable social profile; k = 5 merges the established married couples into their neighbours. **Table 5** shows the six profiles with their departure rates, again fully external to the clustering features.

Table 5 : Entry-profile clusters (k = 6), ordered by departure rate (298,915 persons)

Entry profile	Share	Departure
New EU/EFTA arrivals	24.8%	65.9%
New third-country arrivals	13.8%	63.4%
Single Swiss residents	25.6%	48.0%
Families with young children	17.7%	38.9%
Established married couples	11.7%	34.1%
Single-parent families	6.3%	33.7%

The departure gradient runs from two thirds among new EU/EFTA and third-country arrivals to one third among established couples and single-parent families, with single Swiss residents exactly in between. The typology thus recovers, from twelve entry-time attributes and without ever seeing the outcome, the same structure that the supervised model exploits: who you are when you arrive says a great deal about whether you will stay, and family anchoring in particular retains, whatever the nationality.

5.4 Where the profiles live

Crossing the entry profiles with the district of the first dwelling connects the two deliverables to the geography of **Figure 6**. Nine district-profile cells deviate from the city-wide shares by a factor of 1.5 or more (among cells with at least 200 persons), and they form two coherent patterns rather than a scatter. The first is the Zones foraines: new third-country arrivals reach more than twice their city share there (ratio 2.21), while single Swiss residents (0.49) and single-parent families (0.41) are strongly under-represented. This resolves the anomalous departure rate of that technical district in **Figure 6** : it houses recently arrived residents in collective and institutional settings, precisely the profiles that leave most. The second pattern opposes family districts to arrival districts: families with young children are twice over-represented in Sauvabelin (ratio 1.98), where single Swiss residents are scarce (0.67); established married couples concentrate in Beaulieu/Grey/Boisy (1.68), from which both arrival profiles are largely absent (EU/EFTA 0.58, third-country 0.52); and Bosson/Blécherette similarly under-hosts EU/EFTA arrivals (0.63). None of these facts is visible in the departure map alone or in the profiles alone; it is their crossing that turns a technical district's outlier rate into an interpretable population fact.

6 Discussion

6.1 What the prediction ceiling means

The best honest model reaches an AUC of 0.782, and I read this ceiling as a finding rather than a shortcoming. The features feeding it are unusually rich for entry-time information: a full administrative register, not a survey. If departure were mainly determined by who people are when they arrive, this design would show it. Instead, about five of the twenty points separating the model from the entry-type floor turn out to be calendar exposure, and the attribute-only signal settles near 0.73. The rest of the outcome is decided later, by events that entry-time data cannot see: a job found or lost, a couple formed or separated, a child, a cheaper flat in a neighbouring municipality. The contaminated model makes the same point from the other side: the information that predicts departure almost perfectly does exist in the register, but only in the trajectory as it unfolds, which is to say only once the answer is already being written. Individual departure from a city is, at entry, structurally underdetermined, and the honest question is not whether one can predict who will leave, but how much of the outcome is already inscribed in the starting conditions. The answer here is: a meaningful part, but a minority.

6.2 Uncertainty and robustness

The quantitative results rest on several layers whose uncertainty I can state explicitly. At the data layer, 1.0% of observed transitions remain unexplained after the rupture classification of Section 2.4; any trajectory-based quantity inherits this as an error floor. The reintegration of the 39,853 sedentary snapshot residents as certain non-departures is an assumption, not an observation: it holds if a resident who generates no chainable event has not left, which the register's design supports (a departure generates an event) but cannot strictly prove. At the modelling layer, all reported metrics are stable: five-fold cross-validation within the training set reproduces the test figures to within 0.001, and regenerating the train/test split from a different draw during development moved no comparison and no ranking. At the survival layer, **Figure 4** wears its two biases openly, left truncation for the 2014 stock and window-edge effects for the birth cohort, and the corresponding medians are flagged rather than reported as facts. Deaths are treated as censoring throughout, which slightly overstates survival at high ages; a competing-risks treatment would be the correct refinement.

The typologies deserve their own honesty clause. Silhouette values around 0.25 to 0.30 mean that the clusters overlap: k-means is partitioning a continuum of human trajectories, not discovering separate natural kinds, and the typologies should be read as descriptive summaries whose value is external validation, the steep and coherent departure gradients that the clustering never saw. The selection protocol itself taught a methodological lesson worth recording: with too few random restarts, the stability diagnostic partly measured initialisation luck, and one early verdict reversed under a marginal perturbation of the data. Stability diagnostics are estimators with their own variance; they earned their deciding role in this project only after being computed with thirty restarts per fit.

6.3 Scope and limitations

Three limitations bound the interpretation. First, departure means leaving the municipality of Lausanne: a move to a neighbouring commune of the same urban area counts exactly like a move abroad. Part of the measured departure is therefore metropolitan reshuffling rather than out-migration, and the two cannot be separated with the present data alone; the destination fields of the register would allow it and are the most natural extension. Second, all covariates are frozen at entry by design. This is what makes the prediction honest, but it also means the analysis says nothing about how changes during the stay, a marriage, a new dwelling, a permit upgrade, reshape the propensity to leave; time-varying survival models would address this. Third, the results describe one city, one register, and one window ending in 2026. The pipeline transfers to any Swiss municipality with comparable extracts, but the substantive findings, the profile shares, the district patterns, the departure gradients, are Lausanne's, and the recent-cohort figures in particular are entangled with the post-2020 period they cover.

7 Conclusion and outlook

This project turned eleven years of administrative journals from the Lausanne population register into 493,912 residential episodes for 310,746 persons, with a measured reconstruction uncertainty of 1.0% of transitions, and used them to answer two questions. On prediction: what is known about a person at entry carries real but bounded information about eventual departure, an AUC of 0.782 for the best model, of which a documented part is calendar exposure; the near-perfect performance available from trajectory features is leakage, demonstrated rather than avoided silently. On structure: two independent typologies, five trajectory types and six entry profiles, organise the population along steep departure gradients they were never shown, from short-stay transients (80% departure) to long-settled residents (31%), and from new EU/EFTA arrivals (66%) to anchored families (34%). Crossing the profiles with geography resolved the city's most anomalous district figure. The deliverables are this report, the eight documented notebooks, and the reconstructed episode table itself, which outlives the project as an analytical asset for the statistical office.

Four extensions follow naturally, in increasing order of ambition. Exploiting the destination fields of the register would separate metropolitan reshuffling from genuine out-migration, the main interpretive limit of the present departure definition. Treating deaths as a competing risk rather than censoring would refine the survival estimates at high ages. Time-varying survival models would let covariates evolve during the stay and ask when the propensity to leave changes, not just whose it is. Finally, the reconstructed episodes are exactly the input that sequence analysis methods require [12]; a holistic treatment of full trajectories, rather than the summary features used here, is the natural next step for the typology, and the annual refresh of the register extracts would let all of these run as a maintained observatory rather than a one-off study.

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9 Statement

„Ich erkläre hiermit, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen benutzt habe. Alle Stellen, die wörtlich oder sinngemäss aus Quellen entnommen wurden, habe ich als solche gekennzeichnet. Mir ist bekannt, dass andernfalls die Arbeit als nicht erfüllt bewertet wird und dass die Universitätsleitung bzw. der Senat zum Entzug des aufgrund dieser Arbeit verliehenen Abschlusses bzw. Titels berechtigt ist. Für die Zwecke der Begutachtung und der Überprüfung der Einhaltung der Selbstständigkeitserklärung bzw. der Reglemente betreffend Plagiate erteile ich der Universität Bern das Recht, die dazu erforderlichen Personendaten zu bearbeiten und Nutzungshandlungen vorzunehmen, insbesondere die schriftliche Arbeit zu vervielfältigen und dauerhaft in einer Datenbank zu speichern sowie diese zur Überprüfung von Arbeiten Dritter zu verwenden oder hierzu zur Verfügung zu stellen."

Date: 15.07.2026 Signature(s): Mélisande Rappo

10 Annex A

This annex documents the columns of the two source file types, translated and condensed from the catalogues provided with the extractions. The last column marks the variables actually consumed by the pipeline, as identifiers, chaining inputs, model features, or validation anchors; all other variables were available but not used. Value catalogues for the coded variables (presence status, civil status, permit categories, OFS municipality and country codes, religion) accompany the source catalogues and are available in the project repository.

The monthly files carry a before-and-after column model: variables prefixed MUTATION_ describe the state before the recorded mutation, while their unprefixed counterparts describe the state after it. This pair structure is what allows notebook 5 to verify, at every move, that the dwelling a person leaves matches the dwelling of the previous episode. A small number of delivered columns (notably the family nucleus identifiers of the snapshot file) do not appear in the generic catalogue and were documented from the file itself.

10.1 Monthly mutation files (66 columns)

TABLE A6 : COLUMNS OF THE MONTHLY MUTATION JOURNALS. MUTATION_-PREFIXED VARIABLES CARRY THE PRE-MUTATION STATE.

Column	Description	Used
Event identification		
`DATE_EFFECTIVE`	Effective date of the mutation (date the event took effect)	✓
`MUTATION_DATE`	Administrative entry date of the mutation (date the clerk recorded it)	✓
`CODE_MUTATION`	Mutation type (arrival, departure, internal move, correction, cancellation, ...)	✓
Person identifiers		
`NOAVS`	Swiss social security (AVS) number; replaced by the pseudonymised key, never used	
`NOREFCH`	Population register reference number; basis of the pseudonymised project identifier	✓
`IDHAB`	Cantonal register identifier	
Person attributes		
`CODETA_C`	Presence status of the resident (active, inactive, ...)	✓
`DATNAIS`	Date of birth	✓
`PAYSNAIS`	Country of birth (OFS code)	
`COMNAIS`	Municipality of birth (OFS code)	
`CODE_SEXE`	Sex	✓
`ETAT_CIVIL`	Civil status	✓
`DETATCIVIL`	Date the current civil status took effect	
`DDECES`	Date of death	✓
`NATION`	Nationality (OFS country code)	✓
`TYPERM_C`	Residence permit type (foreign nationals)	✓
`PERMIS_C`	Residence permit category (foreign nationals)	✓
`DATEENTCH`	Date of entry into Switzerland	✓
Arrival, provenance and moves		

Column	Description	Used
DATARR	Date of establishment in Lausanne (latest arrival)	✓
DATDEM	Date of the last move within Lausanne	✓
COMPROV	Municipality of provenance before establishment in Lausanne (OFS code)	✓
COMPROV_C	Municipality of provenance (name)	
PAYSPROVN	Country of provenance (OFS code)	✓
PAYSPROVC	Country of provenance (name)	
MUTATION_COMPROV	Municipality of provenance before the mutation (OFS code)	
MUTATION_COMPROV_C	Municipality of provenance before the mutation (name)	
MUTATION_PAYSPROVN	Country of provenance before the mutation (OFS code)	
MUTATION_PAYSPROVC	Country of provenance before the mutation (name)	
Departure and destination		
TYPE_ADRESSE	Flag set to 'depart' on departure records	
DATDEP	Date of departure from Lausanne	✓
COMDEST	Municipality of destination on departure (OFS code)	
COMDEST_C	Municipality of destination (name)	
PAYSDEST	Country of destination on departure (OFS code)	
PAYSDEST_C	Country of destination (name)	
Secondary residence		
TYPAD_C	Address type: main or secondary residence	✓
XFERESID	Not documented in the source catalogue	
AUTRECOM	Other Swiss municipality of residence (main if Lausanne is secondary, and conversely)	
AUTRPAYS	Country of main residence when Lausanne is a secondary residence	
Current dwelling and address (state after the mutation)		
EGID	Federal building identifier (RegBL)	✓
MENAGE_EWID	Federal dwelling identifier within the building (RegBL)	✓
NOMBRE_PIECE	Number of rooms of the dwelling	✓
SURFACE	Surface of the dwelling (m2)	✓
NOCOM	OFS municipality number of the address	✓
CODRUE	Street code	
NUMERO_ENTREE	Street number	
LIBADRES_C	Address (text)	
NOPOST_C	Postal code	
LOCALITE_C	Locality	
TYPADRES_C	Address type of the current address (main or secondary)	✓
State before the mutation (MUTATION_ prefix)		
MUTATION_EGID	Building identifier before the mutation	✓
MUTATION_EWID	Dwelling identifier before the mutation	✓
MUTATION_PIECES	Rooms of the previous dwelling	
MUTATION_SURFACE	Surface of the previous dwelling	
MUTATION_CODRUE	Street code before the mutation	
MUTATION_NOCOM	OFS municipality number before the mutation	✓
NOIMM_C	Street number before the mutation	
MUTATION_LIBADRES_C	Address before the mutation (text)	
MUTATION_NOPOST_C	Postal code before the mutation	

Column	Description	Used
MUTATION_LOCALITE_C	Locality before the mutation	
MUTATION_TYPADRES_C	Address type before the mutation (main or secondary)	✓
Household		
RM_ID_MENAGE	Household identifier (register-internal)	
TYPE_MENAGE	Household type (private, collective, administrative, out of municipality)	✓
NUMERO_MENAGE	Household identifier (reference version)	✓
PERSMEN	Number of persons in the household	✓
MUTATION_TYPEMENAGE	Household type before the mutation	✓
MUTATION_NUMERO_MENAGE	Household identifier before the mutation	✓
MUTATION_PERSMEN	Household size before the mutation	

10.2 Year-end 2014 snapshot

Table A7 : Columns of the BDCH 2014 snapshot. Derived variables (ages, durations) are recomputed in the pipeline and therefore not consumed from the file.

Column	Description	Used
File and person identifiers		
Annee	Extraction date of the file (31.12.2014)	
Noavs	Swiss social security (AVS) number; never used	
Norefch	Population register reference number; joins the snapshot to the event table	✓
Person attributes		
Codetat	Presence status of the resident	✓
Datnais	Date of birth	✓
AgeCivil	Age at extraction date (derived)	
AgeScol	Theoretical age at the next school year start (derived)	
AgeQuin	Five-year age class (derived)	
AgeGrandClas	Broad age class (derived)	
AgeARLO	ARLO age class (derived)	
PaysnaisCo	Country of birth (OFS code)	✓
PaysnaisCa	Country of birth (name)	
ComnaisCo	Municipality of birth (OFS code)	✓
ComnaisCa	Municipality of birth (name)	
Sexe	Sex	✓
Etatcivil	Civil status	✓
NationCo	Nationality (OFS code)	✓
NationCa	Nationality (name)	
Nation (CH/Etr)	Swiss or foreign national (derived)	
Permis	Residence permit (foreign nationals)	✓
DatentCH	Date of entry into Switzerland	✓
DureeCH	Years lived in Switzerland at extraction (derived)	
Residence in Lausanne		
Domicile	Residence status: main or secondary	✓
DatarrLS	Date of establishment in Lausanne (latest arrival); seeds episode 0	✓

Column	Description	Used
`DureeLS`	Years lived in Lausanne at extraction (derived); kept for validation	✓
`PaysprovCo`	Country of provenance (OFS code)	✓
`PaysprovCa`	Country of provenance (name)	
`ComprovCo`	Municipality of provenance (OFS code)	✓
`ComprovCa`	Municipality of provenance (name)	
`Datdem`	Date of the last move within Lausanne	✓
`Dureedem`	Years lived in the current dwelling (derived)	
Dwelling and address		
`Egid`	Federal building identifier (RegBL)	✓
`Ewid`	Federal dwelling identifier (RegBL)	✓
`NocomOFS`	OFS municipality number	✓
`AdresseCa`	Address (text)	
`Pièces`	Number of rooms of the dwelling (cantonal building register)	✓
`Surface`	Surface of the dwelling (cantonal building register)	✓
Household		
`Typmen`	Household typology (binary in the 2014 schema: private or not)	✓
`Idmen`	Household identifier	✓
Context and geography		
`Religion`	Religion	
`Ilot`	City block number	
`Secteur`	Statistical sector	
`Quartier`	Statistical district	
`GroupBli`	'BLI' sector grouping	
`GroupOAES`	Not documented in the source catalogue	
`CoordX`	X coordinate of the address	
`CoordY`	Y coordinate of the address	
`PGA`	Main land-use zone type (municipal land-use plan)	
`NbVal`	Not documented in the source catalogue	
`NouvArriv`	Not documented in the source catalogue	
`EP / EPS`	Not documented in the source catalogue	

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